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Maiman et al.

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(54) **METHODS AND ARRANGEMENTS FOR A PAYMENT INSTRUMENT WITH A UNIQUE DESIGN**

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(51) **Int. Cl.**

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G06Q 20/20	(2012.01)
G06Q 20/34	(2012.01)
G06Q 20/40	(2012.01)

(52) **U.S. Cl.**

CPC ***G06Q 20/355*** (2013.01); ***G06Q 20/20*** (2013.01); ***G06Q 20/4016*** (2013.01)

(58) **Field of Classification Search**

CPC .. G06Q 20/355; G06Q 20/20; G06Q 20/4016; G06Q 20/204; G06Q 20/34; G06Q 20/409; G06Q 50/04; G06K 19/18

See application file for complete search history.

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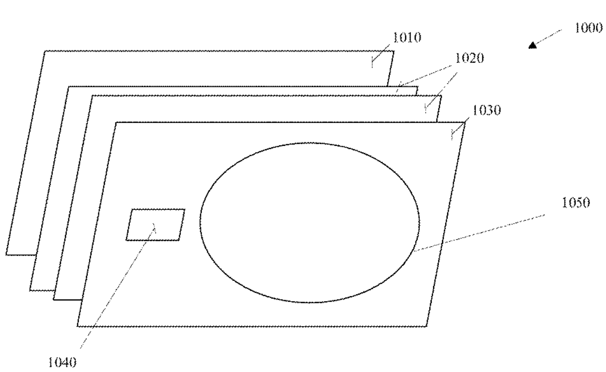
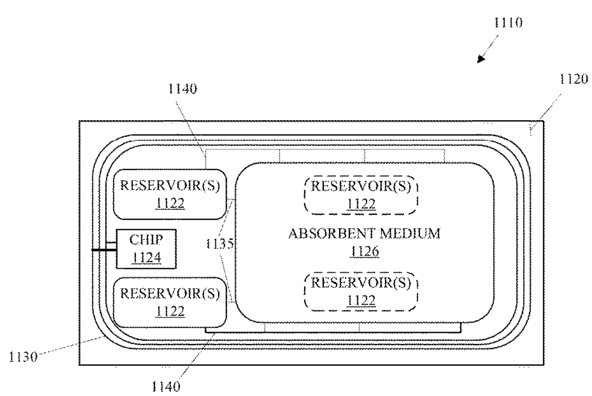
(74) *Attorney, Agent, or Firm* — KDW Firm PLLC

(57) **ABSTRACT**

A payment instrument comprising a bottom layer; one or more intermediate layers, wherein at least one of the one or more intermediate layers comprises: a chip comprising a processor and memory; an antenna coupled with the chip; an absorbent medium; and one or more reservoirs comprising beads, each of the beads comprising dyes of one or more colors, the beads configured to release the dyes onto the absorption medium. The dyes cure on the absorption medium at a rate based on a curing agent applied to the absorbent medium or included in beads of the reservoirs. The payment instrument may also comprise a top layer having at least a partially translucent portion above the absorption medium, wherein the absorption medium is at least partially visible through the at least one partially translucent portion.

20 Claims, 9 Drawing Sheets





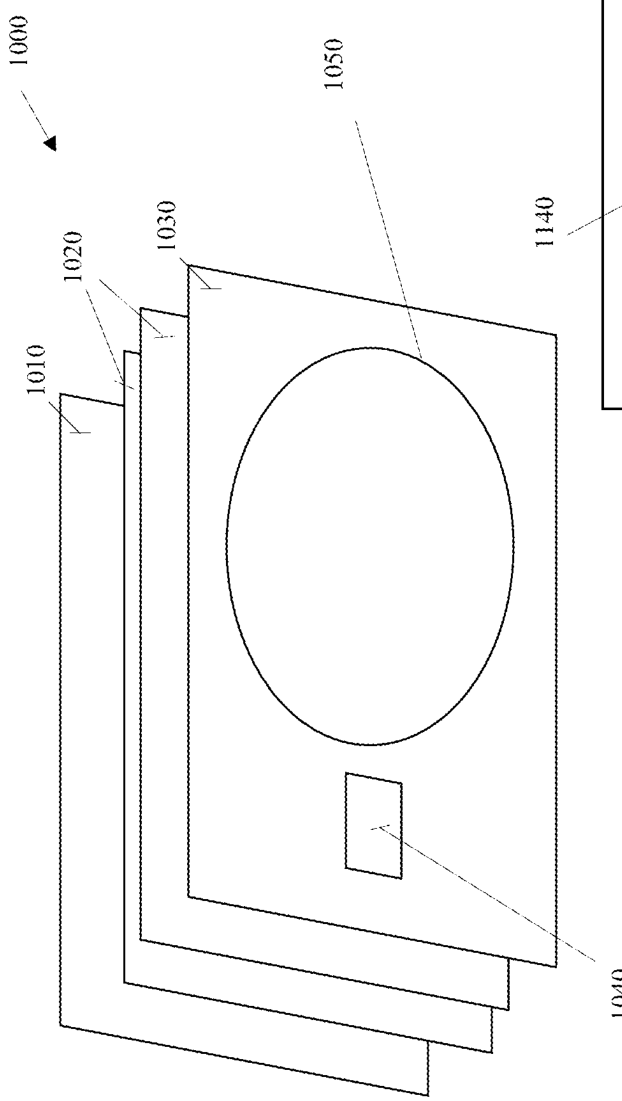

1000
1010
1020
1030
1050
1040

FIG. 1A

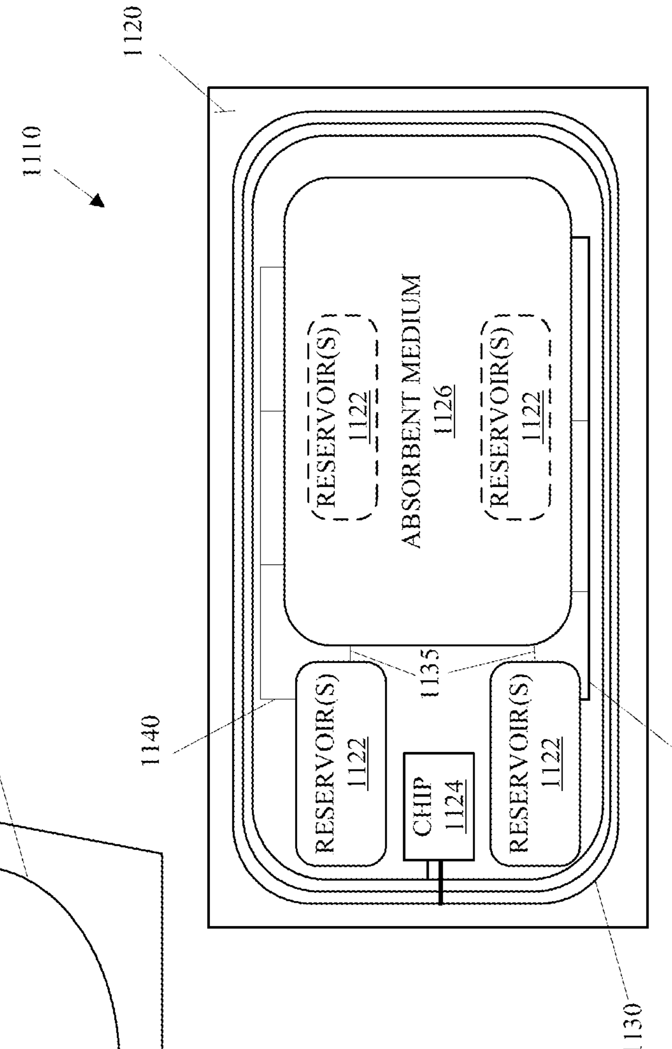

1110
1120
RESERVOIR(S)
1122
ABSORBENT MEDIUM
1126
RESERVOIR(S)
1122
1135
RESERVOIR(S)
1122
CHIP
1124
RESERVOIR(S)
1122
1140
1140
1130

FIG. 1B

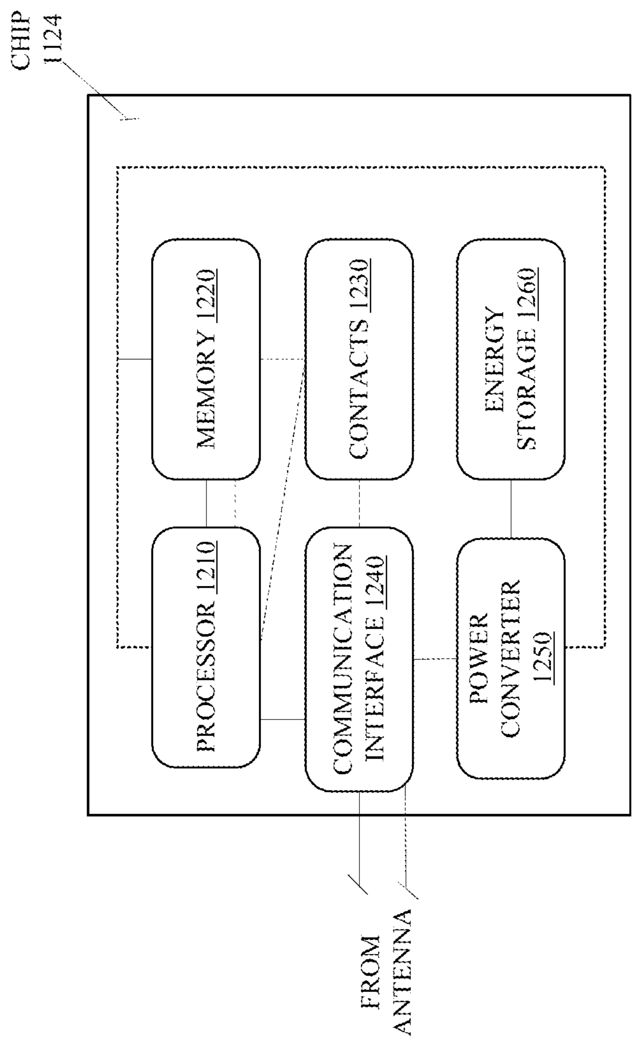

CHIP
1124
MEMORY 1220
CONTACTS 1230
ENERGY
STORAGE 1260
PROCESSOR 1210
COMMUNICATION
INTERFACE 1240
POWER
CONVERTER
1250
FROM
ANTENNA

FIG. 1C

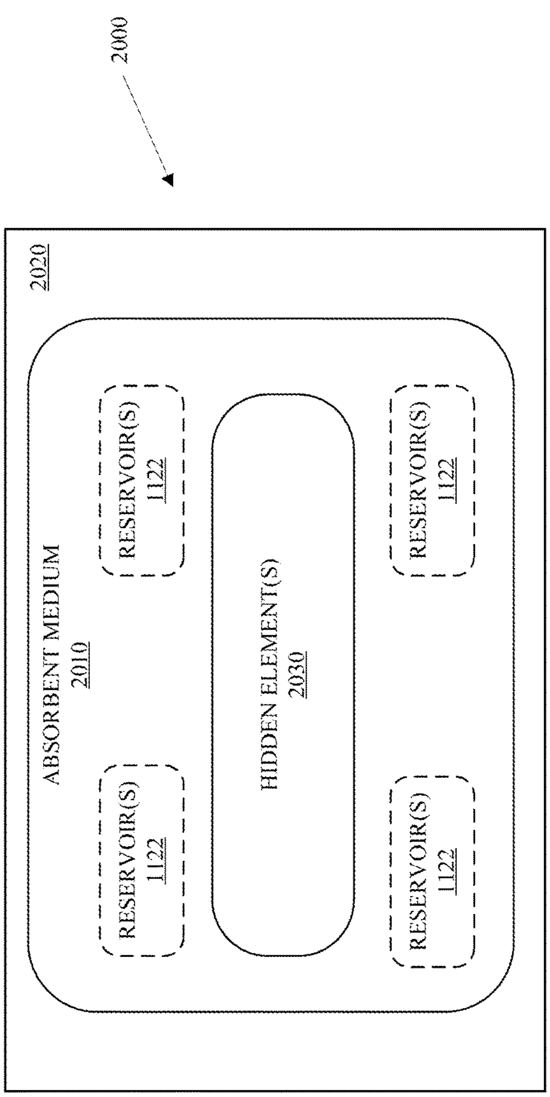

2000
2020
ABSORBENT MEDIUM
2010
RESERVOIR(S)
1122
RESERVOIR(S)
1122
HIDDEN ELEMENT(S)
2030
RESERVOIR(S)
1122
RESERVOIR(S)
1122

FIG. 2A

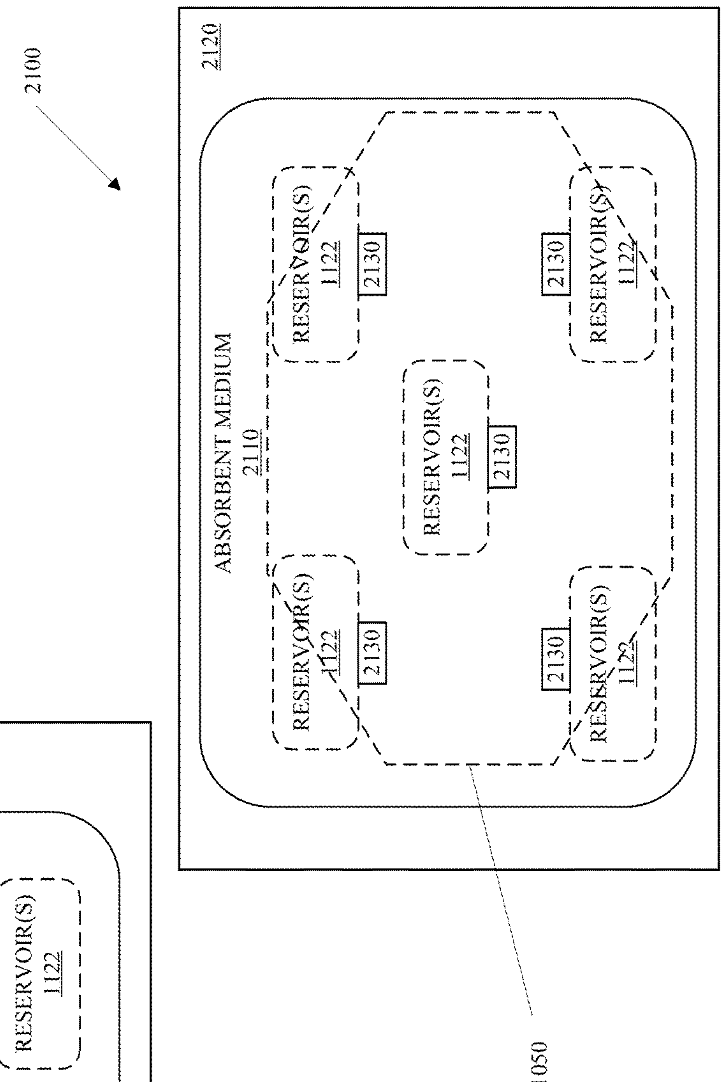

2100
2120
ABSORBENT MEDIUM
2110
RESERVOIR(S)
1122
2130
RESERVOIR(S)
1122
2130
RESERVOIR(S)
1122
2130
RESERVOIR(S)
1122
2130
RESERVOIR(S)
1122
2130
1050

FIG. 2B

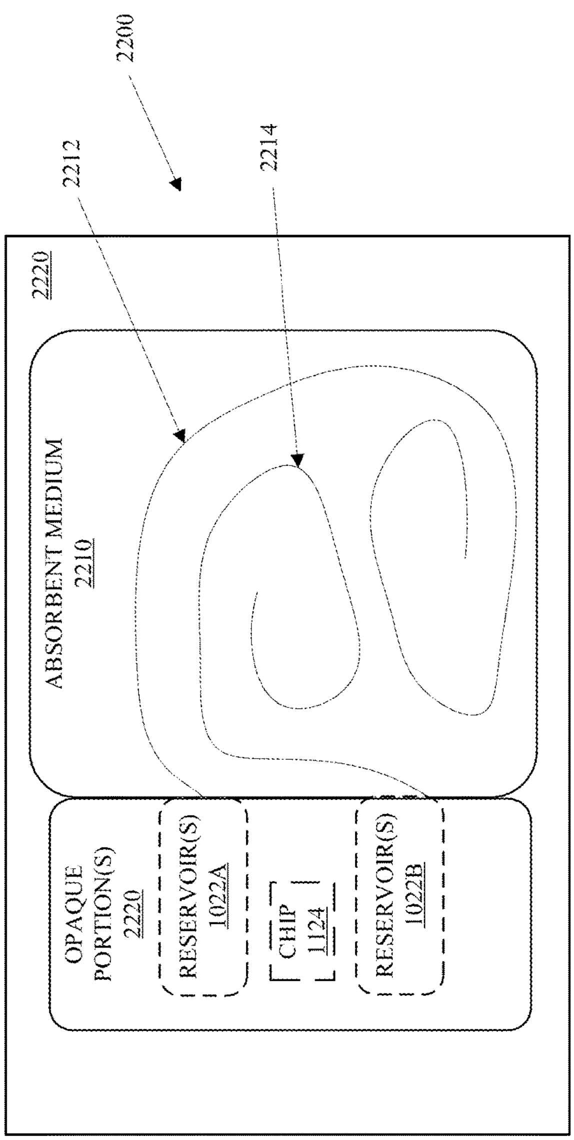

2200
2212
2214
2220
ABSORBENT MEDIUM
2210
OPAQUE PORTION(S)
2220
RESERVOIR(S)
1022A
CHIP
1124
RESERVOIR(S)
1022B

FIG. 2C

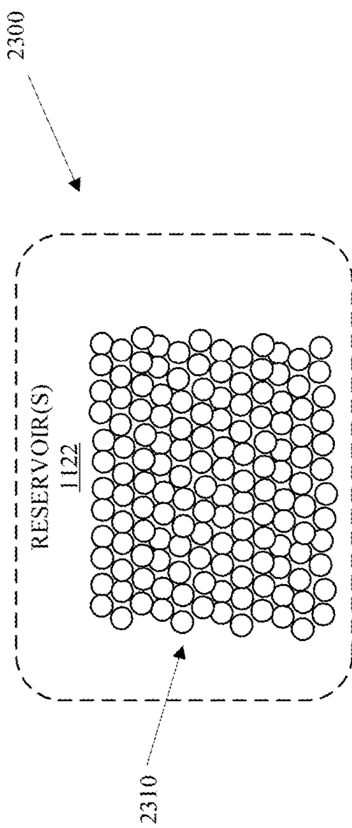

2300
RESERVOIR(S)
1122
2310

FIG. 2D

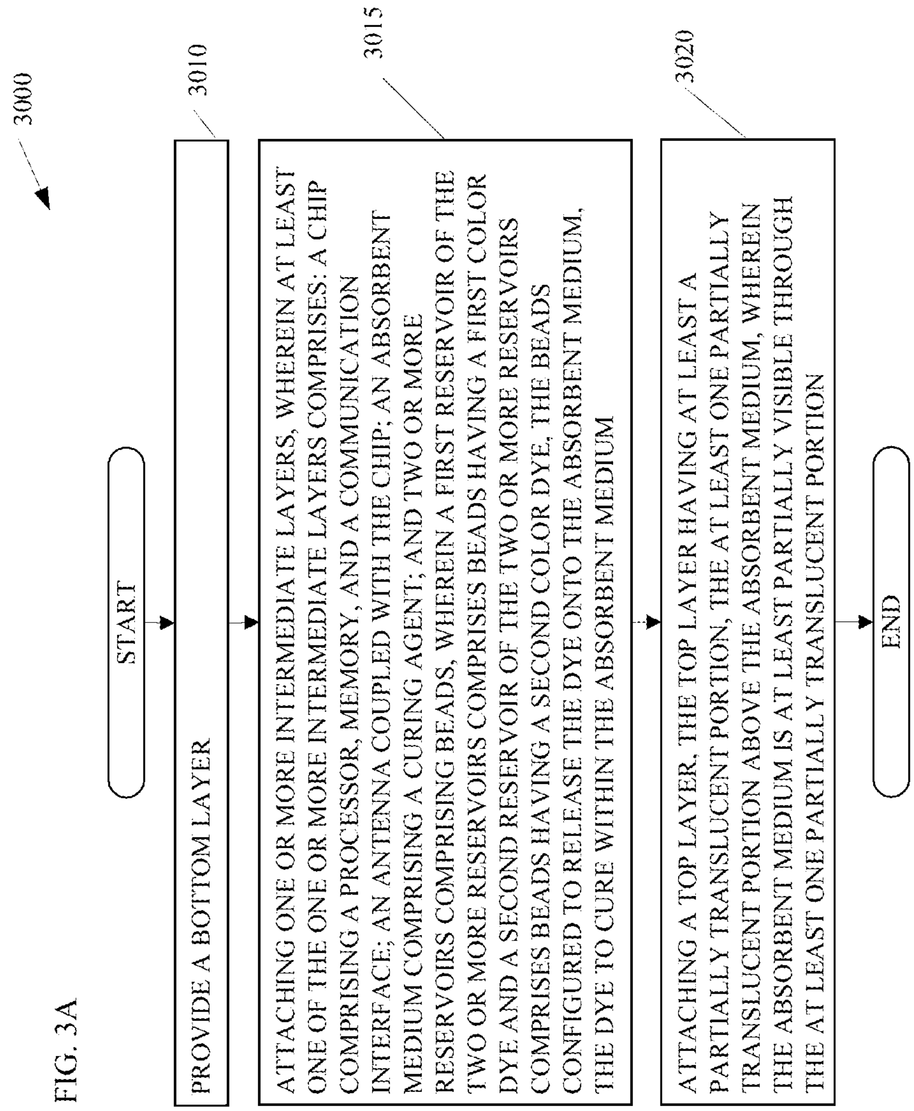

3000
START
3010
PROVIDE A BOTTOM LAYER
3015
ATTACHING ONE OR MORE INTERMEDIATE LAYERS, WHEREIN AT LEAST ONE OF THE ONE OR MORE INTERMEDIATE LAYERS COMPRISES: A CHIP COMPRISING A PROCESSOR, MEMORY, AND A COMMUNICATION INTERFACE; AN ANTENNA COUPLED WITH THE CHIP; AN ABSORBENT MEDIUM COMPRISING A CURING AGENT; AND TWO OR MORE RESERVOIRS COMPRISING BEADS, WHEREIN A FIRST RESERVOIR OF THE TWO OR MORE RESERVOIRS COMPRISES BEADS HAVING A FIRST COLOR DYE AND A SECOND RESERVOIR OF THE TWO OR MORE RESERVOIRS COMPRISES BEADS HAVING A SECOND COLOR DYE, THE BEADS CONFIGURED TO RELEASE THE DYE ONTO THE ABSORBENT MEDIUM, THE DYE TO CURE WITHIN THE ABSORBENT MEDIUM
3020
ATTACHING A TOP LAYER, THE TOP LAYER HAVING AT LEAST A PARTIALLY TRANSLUCENT PORTION, THE AT LEAST ONE PARTIALLY TRANSLUCENT PORTION ABOVE THE ABSORBENT MEDIUM, WHEREIN THE ABSORBENT MEDIUM IS AT LEAST PARTIALLY VISIBLE THROUGH THE AT LEAST ONE PARTIALLY TRANSLUCENT PORTION
END

FIG. 3A

FIG. 3B

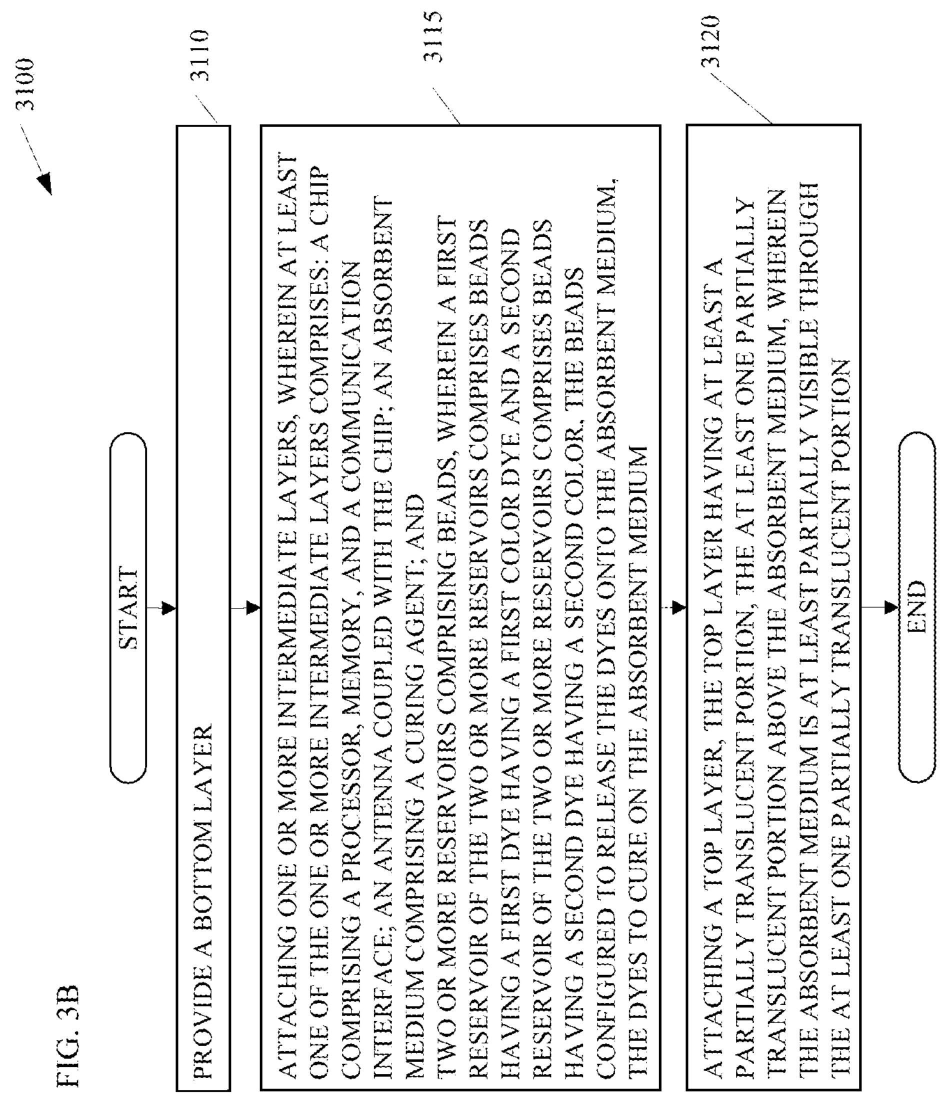

3100
START
3110
PROVIDE A BOTTOM LAYER
3115
ATTACHING ONE OR MORE INTERMEDIATE LAYERS, WHEREIN AT LEAST ONE OF THE ONE OR MORE INTERMEDIATE LAYERS COMPRISES: A CHIP COMPRISING A PROCESSOR, MEMORY, AND A COMMUNICATION INTERFACE; AN ANTENNA COUPLED WITH THE CHIP; AN ABSORBENT MEDIUM COMPRISING A CURING AGENT; AND TWO OR MORE RESERVOIRS COMPRISING BEADS, WHEREIN A FIRST RESERVOIR OF THE TWO OR MORE RESERVOIRS COMPRISES BEADS HAVING A FIRST DYE HAVING A FIRST COLOR DYE AND A SECOND RESERVOIR OF THE TWO OR MORE RESERVOIRS COMPRISES BEADS HAVING A SECOND DYE HAVING A SECOND COLOR, THE BEADS CONFIGURED TO RELEASE THE DYES ONTO THE ABSORBENT MEDIUM, THE DYES TO CURE ON THE ABSORBENT MEDIUM
3120
ATTACHING A TOP LAYER, THE TOP LAYER HAVING AT LEAST A PARTIALLY TRANSLUCENT PORTION, THE AT LEAST ONE PARTIALLY TRANSLUCENT PORTION ABOVE THE ABSORBENT MEDIUM, WHEREIN THE ABSORBENT MEDIUM IS AT LEAST PARTIALLY VISIBLE THROUGH THE AT LEAST ONE PARTIALLY TRANSLUCENT PORTION
END

FIG. 3C

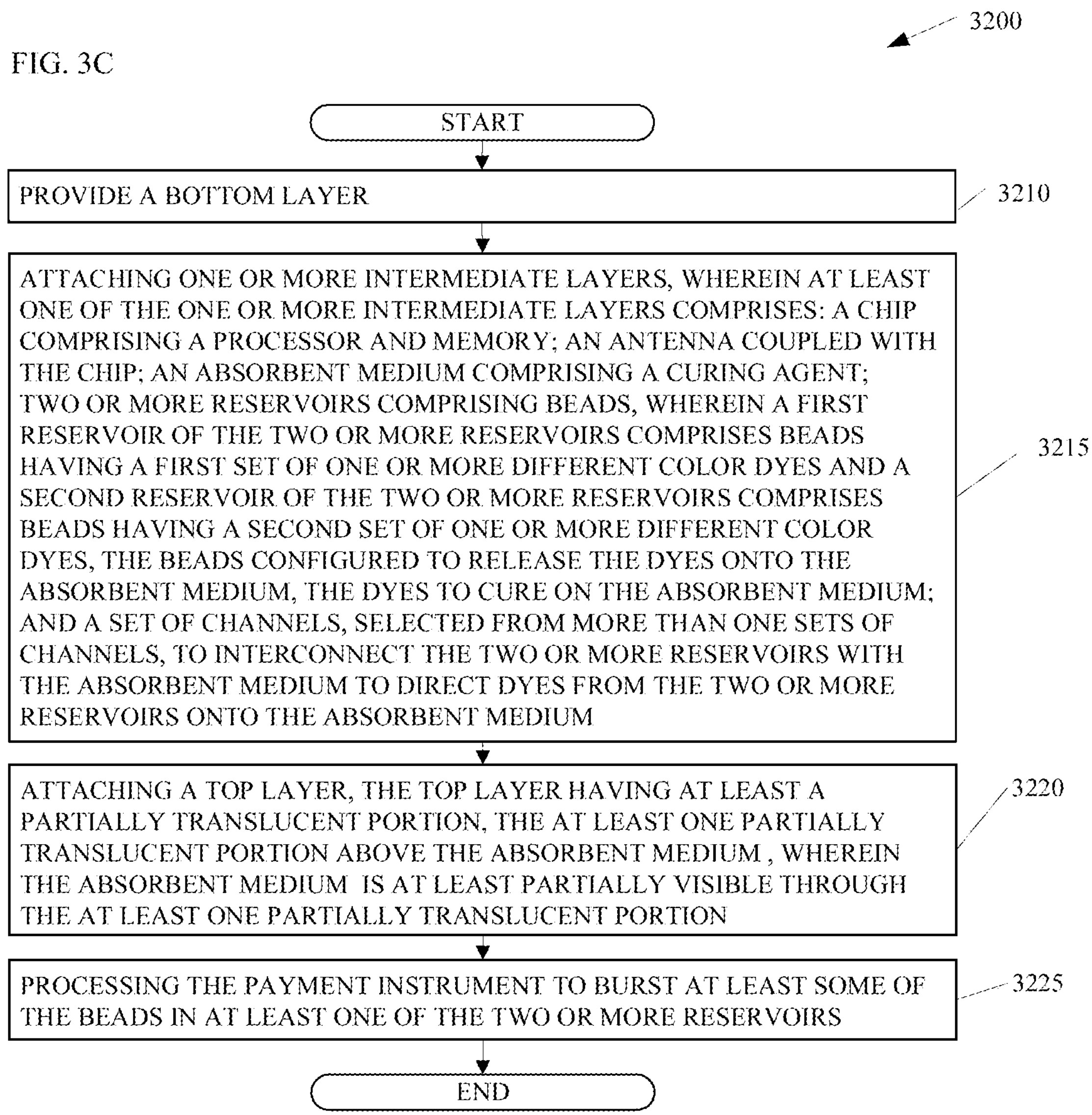

3200
START
PROVIDE A BOTTOM LAYER
3210
ATTACHING ONE OR MORE INTERMEDIATE LAYERS, WHEREIN AT LEAST ONE OF THE ONE OR MORE INTERMEDIATE LAYERS COMPRISES: A CHIP COMPRISING A PROCESSOR AND MEMORY; AN ANTENNA COUPLED WITH THE CHIP; AN ABSORBENT MEDIUM COMPRISING A CURING AGENT; TWO OR MORE RESERVOIRS COMPRISING BEADS, WHEREIN A FIRST RESERVOIR OF THE TWO OR MORE RESERVOIRS COMPRISES BEADS HAVING A FIRST SET OF ONE OR MORE DIFFERENT COLOR DYES AND A SECOND RESERVOIR OF THE TWO OR MORE RESERVOIRS COMPRISES BEADS HAVING A SECOND SET OF ONE OR MORE DIFFERENT COLOR DYES, THE BEADS CONFIGURED TO RELEASE THE DYES ONTO THE ABSORBENT MEDIUM, THE DYES TO CURE ON THE ABSORBENT MEDIUM; AND A SET OF CHANNELS, SELECTED FROM MORE THAN ONE SETS OF CHANNELS, TO INTERCONNECT THE TWO OR MORE RESERVOIRS WITH THE ABSORBENT MEDIUM TO DIRECT DYES FROM THE TWO OR MORE RESERVOIRS ONTO THE ABSORBENT MEDIUM
3215
ATTACHING A TOP LAYER, THE TOP LAYER HAVING AT LEAST A PARTIALLY TRANSLUCENT PORTION, THE AT LEAST ONE PARTIALLY TRANSLUCENT PORTION ABOVE THE ABSORBENT MEDIUM , WHEREIN THE ABSORBENT MEDIUM IS AT LEAST PARTIALLY VISIBLE THROUGH THE AT LEAST ONE PARTIALLY TRANSLUCENT PORTION
3220
PROCESSING THE PAYMENT INSTRUMENT TO BURST AT LEAST SOME OF THE BEADS IN AT LEAST ONE OF THE TWO OR MORE RESERVOIRS
3225
END

FIG. 4

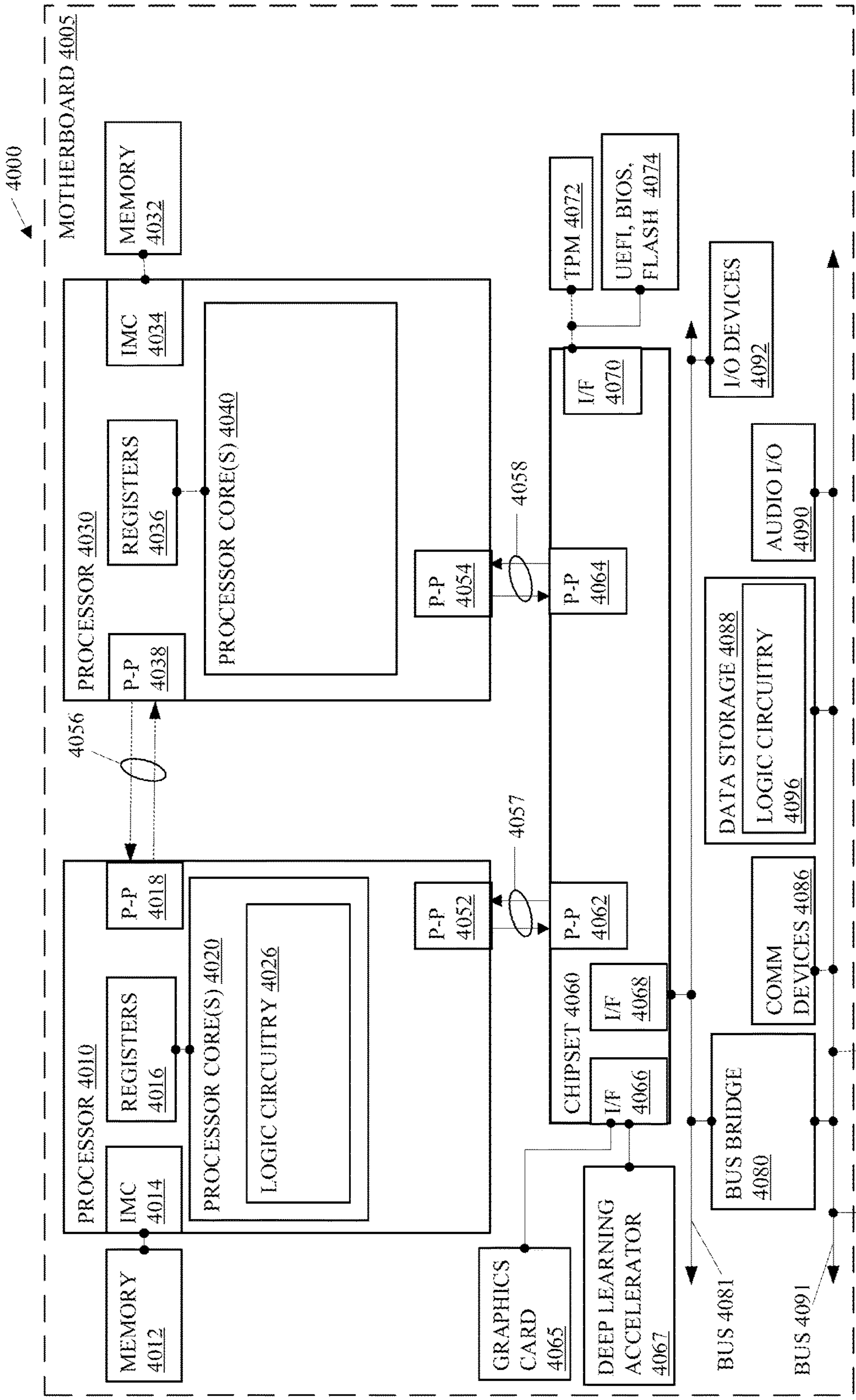

4000
MOTHERBOARD 4005
MEMORY 4032
IMC 4034
REGISTERS 4036
PROCESSOR CORE(S) 4040
PROCESSOR 4030
P-P 4038
P-P 4054
4058
P-P 4064
TPM 4072
UEFI, BIOS, FLASH 4074
I/F 4070
I/O DEVICES 4092
AUDIO I/O 4090
DATA STORAGE 4088
LOGIC CIRCUITRY 4096
4056
4057
PROCESSOR 4010
P-P 4018
PROCESSOR CORE(S) 4020
LOGIC CIRCUITRY 4026
REGISTERS 4016
IMC 4014
MEMORY 4012
P-P 4052
P-P 4062
CHIPSET 4060
I/F 4068
I/F 4066
COMM DEVICES 4086
BUS BRIDGE 4080
MOUSE 4084
KEYBOARD 4082
GRAPHICS CARD 4065
DEEP LEARNING ACCELERATOR 4067
BUS 4081
BUS 4091

FIG. 5

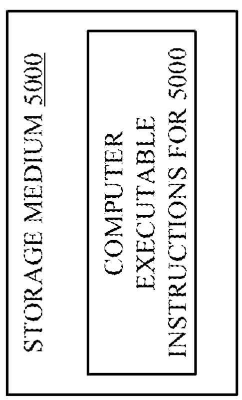
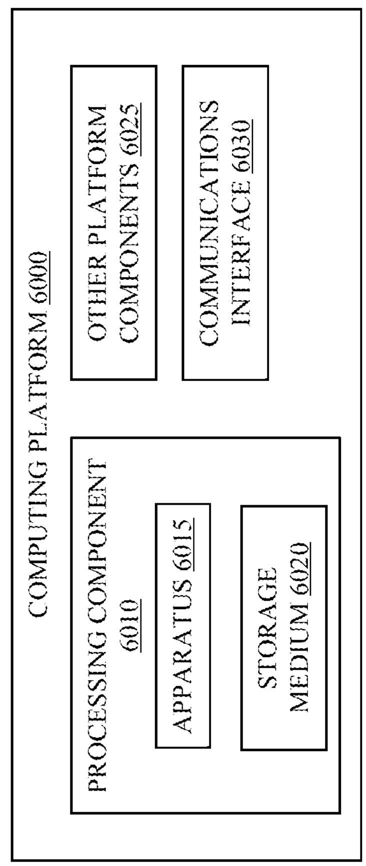

STORAGE MEDIUM 5000
COMPUTER EXECUTABLE INSTRUCTIONS FOR 5000

FIG. 6


COMPUTING PLATFORM 6000
PROCESSING COMPONENT 6010
APPARATUS 6015
STORAGE MEDIUM 6020
OTHER PLATFORM COMPONENTS 6025
COMMUNICATIONS INTERFACE 6030

METHODS AND ARRANGEMENTS FOR A PAYMENT INSTRUMENT WITH A UNIQUE DESIGN

RELATED APPLICATION

This application is a continuation application of U.S. patent application Ser. No. 17/984,467, filed on Nov. 10, 2022, entitled "METHODS AND ARRANGEMENTS FOR A PAYMENT INSTRUMENT WITH A UNIQUE DESIGN". The contents of the aforementioned application is incorporated herein by reference.

TECHNICAL FIELD

Embodiments described herein are in the field of payment instrument design. More particularly, the embodiments relate to methods and arrangements for a payment instrument with a unique design for each customer.

BACKGROUND

A modern credit card has three primary methods of making an in-person transaction with a merchant: tap, chip, and swipe. Each method relies on different technology embedded within the credit card. These technologies are substantially independent of one another such that one payment method may fail while the other two continue to function.

Consumers tend to carry credit cards from more than one issuing bank for various reasons. Credit cards tend to be generic or include relatively common themes such as pictures, emblems, or trademarks on the face, or top layer, of the credit cards to help to distinguish the credit cards from one another. When making a purchase, however, consumers tend to select the same credit card more often than other credit cards. Furthermore, credit cards with generic designs make it difficult to detect the difference between a counterfeit credit card and a real credit card.

SUMMARY

Embodiments may include methods and arrangements such as methods, devices, apparatuses, systems, storage media, and the like. For example, a first embodiment may include a payment instrument. The payment instrument may comprise a bottom layer; and one or more intermediate layers, wherein at least one of the one or more intermediate layers comprises: a chip comprising a chip comprising a processor, memory, a communication interface, and power converter; an antenna coupled with the chip; an absorbent medium comprising a curing agent; and two or more reservoirs comprising beads. A first reservoir of the two or more reservoirs may comprise beads having a first color dye and a second reservoir of the two or more reservoirs may comprise beads having a second color dye. The beads may be configured to release the dye onto the absorbent medium and the dye may cure within the absorbent medium. The payment instrument may also comprise a top layer. The top layer may have at least a partially translucent portion. The at least one partially translucent portion may be above the absorbent medium, wherein the absorbent medium is at least partially visible through the at least one partially translucent portion.

A second embodiment may include a payment instrument. The payment instrument may comprise a bottom layer and one or more intermediate layers. At least one of the one or more intermediate layers may comprise a chip comprising a chip comprising a processor, memory, a communication interface, and power converter; an antenna coupled with the chip; an absorbent medium comprising a curing agent; and two or more reservoirs comprising beads. A first reservoir of the two or more reservoirs may comprise beads having a first dye having a first color dye and a second reservoir of the two or more reservoirs may comprise beads having a second dye having a second color. The beads may be configured to release the dyes onto the absorbent medium and the dyes may cure on the absorbent medium. The payment instrument may also comprise a top layer, the top layer having at least a partially translucent portion. The at least one partially translucent portion may reside above the absorbent medium so the absorbent medium is at least partially visible through the at least one partially translucent portion.

A third embodiment may include a method to create a payment instrument with a unique design. The method may comprise providing a bottom layer and attaching one or more intermediate layers. At least one of the one or more intermediate layers may comprise a chip comprising a chip comprising a processor, memory, a communication interface, and power converter; an antenna coupled with the chip; an absorbent medium comprising a curing agent; and two or more reservoirs comprising beads. A first reservoir of the two or more reservoirs may comprise beads having a first set of one or more different color dyes and a second reservoir of the two or more reservoirs may comprise beads having a second set of one or more different color dyes. The beads may be configured to release the dyes onto the absorbent medium and the dyes to cure on the absorbent medium. At least one of the one or more intermediate layers may comprise a set of channels, selected from more than one sets of channels, to interconnect the two or more reservoirs with the absorbent medium to direct dyes from the two or more reservoirs onto the absorbent medium. The method may also comprise attaching a top layer. The top layer may have at least a partially translucent portion above the absorbent medium. The absorbent medium may be at least partially visible through the at least one partially translucent portion. The method may further comprise processing the payment instrument to burst at least some of the beads in at least one of the two or more reservoirs.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A depicts an embodiment of a multi-layer payment instrument;

FIG. 1B depicts an embodiment of an intermediate layer, such as the intermediate layer illustrated in FIG. 1A;

FIG. 1C depicts an embodiment of a chip such as the chip illustrated in FIG. 1B;

FIG. 2A depicts an alternative embodiment of an intermediate layer, such as the intermediate layer illustrated in FIG. 1A;

FIG. 2B depicts an alternative embodiment of an intermediate layer, such as the intermediate layer illustrated in FIG. 1A;

FIG. 2C depicts an alternative embodiment of an intermediate layer, such as the intermediate layer illustrated in FIG. 1A;

FIG. 2D depicts an embodiment of a reservoir, such as the reservoirs illustrated in FIG. 1B and 2A-2C; and

FIGS. 3A-C depict embodiments of flowcharts to manufacture payment instruments such as the payment instruments shown in FIGS. 1A-B and 2A-D; and

FIG. 4 depicts an embodiment of a system including a multiple-processor platform, a chipset, buses, and accessories such as a server to perform the processes described in conjunction with FIGS. 3A-C; and

FIGS. 5-6 depict embodiments of a storage medium and a computing platform such as a server to perform the processes described in conjunction with FIGS. 3A-C.

DETAILED DESCRIPTION OF EMBODIMENTS

The following is a detailed description of embodiments depicted in the drawings. The detailed description covers all modifications, equivalents, and alternatives falling within the appended claims.

Customers may begin to rely on one or more payment instruments such as credit cards to perform transactions. Many customers have typical, repetitive, or periodic expenses for which they rely on one or more credit cards. For instance, customers may use one or more credit cards to purchase gas for their vehicles once a week, eat lunch at a restaurant or cafeteria a few times a week, pick up groceries a few times a month, and/or the like. Some customers fall into a routine in which they use the same payment instrument to perform most of the day-to-day transactions because they prefer use of that card for one reason or another.

Many payment instruments are plastic and/or metal credit cards that include payment interfaces of various technologies to conduct transactions for customers. The payment interfaces may include a magnetic stripe including information associated with the customer that a card reader can read to process a transaction. Magnetic stripes contain magnetically stored information for conducting a transaction and are typically applied to credit cards as a hot foil tape. In many embodiments, the credit cards include a high-coercivity magnetic stripe that requires a higher magnetic energy to record (e.g., 4000 oersted (Oe)) than medium-coercivity (e.g., 2750 Oe) and low-coercivity (e.g., 300 Oe).

Payment instruments may also comprise a chip such as a microchip with contacts as a payment interface and such payment instruments are often referred to as Chip and PIN (personal identification number) or Chip and Signature cards. The chip may comprise a processor that has a contacted payment interface. With the Chip and PIN cards, the POS terminal, if it has the corresponding capabilities, may verify the identity of the customer with a PIN via the chip, whereas Chip and Signature cards require a signature to verify the identity of the customer. In some embodiments, the chip may also generate a packet for transmission to the payment instrument issuer such as an encrypted packet with a random number that can verify the operation of the chip and the association of the chip with the customer's account.

Payment instruments may also comprise a contactless payment interface such as a near field communications (NFC) payment interface. In some embodiments, the contactless payment interface may comprise a legacy magnetitic stripe radio frequency identifier (RFID) tag and, in further embodiments, the contactless payment interface may comprise an NFC payment interface coupled with the chip. Either or both of such contactless payment interfaces may include an antenna that is typically embedded in the payment instrument on an intermediate layer of the payment instrument and encircles a portion of the payment instrument. The antenna enables reception of radio signals and for interacting with a tap type payment interface on a POS terminal.

Reception of the radio signals may include radio signals for communications between the processor on the payment instrument and the POS terminal and/or may include radio signals configured to transfer power from the POS terminal to the payment instrument wirelessly.

How consumers choose the credit card that they prefer to use may depend on several factors and one factor may be aesthetics and uniqueness of the design of the credit card. The design may catch the consumer's attention.

When the credit card is lost, stolen, or skimmed, however, the various technologies may not prevent fraudulent transactions with the credit card. For instance, the tap, chip, or swipe may be accepted with a signature or pin that can be faked, observed, or skimmed; with easily determined information such as a zip code; or with other information entered by the user on a terminal that can be monitored, seen, or skimmed. An obviously counterfeit card may catch the attention of a merchant and possibly avoid an attempt to perform fraudulent transactions in certain situations.

Embodiments may comprise a payment instrument and the manufacture of a payment instrument with a unique design. The payment instrument may comprise three or more layers attached for a single payment instrument. The payment instrument may comprise a bottom layer, one or more intermediate layers, and a top layer. The bottom layer may provide a foundation or base for the payment instrument. In some embodiments, the bottom layer may establish a magnitude of rigidity of the payment instrument that may be reinforced or increased by the one or more intermediate layers and the top layer.

The one or more intermediate layers may be attached (e.g., laminated, glued, or otherwise affixed) to the bottom layer. The one or more intermediate layers may comprise a chip comprising a processor, memory, a communication interface, and power converter. The chip may comprise an integrated circuit such as a system-on-a-chip (SoC) or an application-specific integrated circuit (ASIC). In many embodiments, the chip may capture energy from a point-of-sale (POS) terminal through conductive contacts on the top layer of the payment instrument or via one or more antennas coupled with a power converter. When the payment instrument is inserted into a POS terminal, the conductive contacts may contact conductive contacts of the POS terminal to power the circuitry of the chip and to establish a physical communication channel between the payment instrument and the POS terminal.

Alternatively, when the payment instrument is held close to, e.g., a near field communications (NFC) interface of a POS terminal, the POS terminal may wirelessly transfer energy to the payment instrument and a power converter of the payment instrument may collect the energy. The payment instrument may consume power to receive a transmission from the POS terminal and may collect energy in an energy storage until sufficient energy is collected to power a transmission of a response from the chip via a wireless radio of the communication interface.

The one or more intermediate layers may also include an absorbent medium and one or more reservoirs. In some embodiments, the absorbent medium may include a curing agent to cure dyes. The reservoirs may include one or more dyes to disperse onto the absorbent medium during and/or after then manufacture of the payment instrument. For instance, in some embodiments, the user may insert a payment instrument into a POS terminal to process a transaction and the process of insertion may cause a portion of the dyes in one or more of the reservoirs to be released onto the absorbent medium to create a unique design with the absorbent medium.

In some embodiments, the dyes may reside in beads that burst or dissolve. In some embodiments, beads having dyes may burst or dissolve as a result of processing the payment instrument at the time of manufacture of the payment instrument. In some embodiments, some beads may dissolve in response to the presence dye from other burst or dissolved beads. In some embodiments, handling of the payment instrument by a customer may burst some remaining beads.

In further embodiments, the portion of the absorbent medium precoated with a curing agent may form a pattern such as a logo, a code, a QR code, a security feature, a three-dimensional design, a combination thereof, and/or the like. For instance, the absorbent medium may include a pattern that is a security feature, or the security feature may be applied to the absorbent medium and may either remain invisible except under certain light conditions or may appear as a reaction to the application of dyes to the absorbent medium. In such embodiments, the security feature may facilitate detection of counterfeit payment instruments.

In many embodiments, a dye may be placed into a cavity of one or more of the intermediate layers and different color dyes may be placed in different cavities. In some embodiments, the dyes are encapsulated in beads that may be stored in one or more cavities of one or more of the intermediate layers and may be crushed in response to insertion of the payment instrument into a POS terminal, as a result of other processing of the payment instrument during manufacture or may be crushed by a customer during use of the payment instrument. In further embodiments, the beads may dissolve to release the dyes onto the absorbent medium in response to the presence of dye from other beads of another agent included in the cavities or reservoirs with the beads.

In some embodiments, beads having different colors may be combined in a cavity or reservoir to create a combined color or a spectrum of colors related to the combination of the colors of the dyes in the beads as the dyes mix and are absorbed into the absorbent medium. In some embodiments, the beads may include different color dyes having different viscosities and/or absorption rates to cause some colors of dyes to absorb into the absorbent medium at faster rate and some colors of the dyes to absorb into the absorbent medium at slower rates.

In some embodiments, a coating may be placed on the absorbent medium to prevent or attenuate adherence or absorption of the dyes on a portion of the surface of the absorbent medium. In some embodiments, the coating may form a pattern such as a logo, a code, a QR code, a security feature, a three-dimensional feature, a combination thereof, and/or the like.

Turning now to the drawings, FIGS. **1**A-C depict embodiments of payment instruments such as credit cards with contacted and contactless payment interfaces. Some embodiments may also have a magnetic stripe attached to the payment instrument.

FIG. **1**A shows a multi-layer payment instrument **1000**. The multi-layer payment instrument **1100** comprises a bottom layer **1010** or base layer, one or more intermediate layers **1020**, and a top layer **1030**. The bottom layer **1110** may form a foundation for the multi-layer payment instrument **1000** that provides structural integrity and a desired amount of rigidity. The one or more intermediate layers **1020** may include one or more laminations attached to the bottom layer **1010** to add functionality such as a chip **1124**, antennas **1130**, an absorbent medium **1126**, reservoir(s) **1122**, and the like illustrated on an intermediate layer **1120** in FIG. **1**B.

The chip **1124** may comprise one or more layers of printed circuits and/or circuit elements embedded into the intermediate layers. In some embodiments, at least a portion of the chip **1124** may be printed on or embedded into the bottom layer **1010**. Contacts **1040** for the chip reside on the top layer **1030** to physically connect with contacts in a POS terminal to provide a contacted payment interface. Furthermore, the top layer **1030** may include at least a translucent window **1050** to view the absorbent medium **1126** through the top layer **1030**.

In some embodiments, the translucent window **1050** may be transparent. In other embodiments, the translucent window **1050** may comprise a tint or have a partial opacity to filter the appearance of the absorbent medium **1126**.

The antennas **1130** may form one or more loops of conductive material printed on the intermediate layer **1120** about the multi-layer payment instrument **1000**. In some embodiments, the one or more loops of conductive material may be printed in different sizes to capture different types of signals such as near field communications (NFC) or wireless power transfer. In some embodiments, the chip **1124** may select the antenna configuration based on the functionality in use.

In some embodiments, the absorbent medium **1126** may comprise a cloth such as a cotton, paper, and/or the like. The absorbent medium **1126** may comprise one or more pieces of cloth coated with a curing agent to sure the dyes in the reservoir(s) **1122**, one or cloth coated with an agent to prevent or reduce absorption of the dyes in the reservoir(s) **1122**, a combination thereof, and/or the like. In some embodiments, a security feature of the payment instrument may reside in a pattern of the one or more pieces of cloth coated with the curing agent, coated with an agent to reduce or prevent absorption, or a combination thereof.

The reservoir(s) **1122** may comprise one or more cavities to contain the dyes. The cavities may be located at various locations about the absorbent medium **1126** to disperse the dyes in patterns that may be dependent upon specific usage of a payment instrument **1000**. For instance, FIG. **1**B illustrates two reservoir(s) **1122** located adjacent to the chip **1124**. When the payment instrument **1000** is inserted into a POS terminal, the contact with the POS terminal may cause the reservoir(s) **1122** adjacent to the chip **1124** to release some of the dyes on to the absorbent medium **1126**. In some embodiments, the reservoir(s) **1122** may disperse the dyes directly on to the absorbent medium **1126** through a channel **1135** with a single connection interconnecting the absorbent medium **1126** with the reservoir(s) **1122**. The channel may be between a reservoir on a layer underneath the absorbent medium **1126** up to the absorbent medium **1126** and/or may include a channel directly from the reservoir on the same intermediate layer **1120** as the absorbent medium **1126**. In other embodiments, the reservoir(s) **1122** may couple with a channel with multiple connections **1140** to the absorbent medium **1126**. In further embodiments, the reservoir(s) **1122** may couple with a channel having a pattern such as a curved pattern or a spiral pattern to disperse the dyes in the curved pattern or spiral pattern about the absorbent medium **1126**.

In some embodiments, the absorbent medium **1126** comprises a base color that combines with the color of a dye to create a different color. In some embodiments, the absorbent medium **1126** comprises a pattern of base colors that combine with the color of a dye to create different colors. In some embodiments, the absorbent medium **1126** comprises a pattern of base colors that combine with multiple dyes of different colors, wherein each dye absorbed into a portion of the absorbent medium **1126** changes the color of that portion of the absorbent medium **1126**.

In some embodiments, the dyes reside in beads. The beads may have a coating that dissolves in response to dyes within and external to the beads and/or ruptures when pressure is applied to the reservoir(s) **1122**. For instance, a customer may press on the payment instrument **1000** firmly on one of the reservoir(s) **1122** and that pressure may burst some of the beads, causing the dyes to contact the surface of the absorbent medium **1126**. The absorbent medium **1126** may absorb the dyes at rates relative to the viscosity or absorbency properties of the dyes and change in appearance based on the curing time of the dyes. For instance, greater the quantities or concentrations of the curing agent may cause the dyes to cure faster on the absorbent medium **1126** in some embodiments. In some embodiments, the curing time of a dye may affect the depth of the color in the absorbent medium **1126**. For instance, a faster curing time for a dye may reduce the depth to which the color is absorbed into the absorption medium **1126**. A slower curing time may increase the depth to which the color is absorbed into the absorption medium **1126**, varying the perception of the color of the dye through the translucent window **1050** of the top layer **1030**.

In some embodiments, some of the reservoir(s) **1122** may include a first agent to hasten the pace of curing of the dyes on the absorbent medium **1126**. And other reservoir(s) **1122** in the payment instrument **1000** may include a second agent to prevent or slow the absorption rate of the dyes into the absorbent medium **1126**.

FIG. **1**C depicts an embodiment of a chip **1124** such as the chip **1124** illustrated in FIG. **1**B. The chip **1124** may comprise an integrated circuit built via one or more layers of printed or embedded circuit patterns painted or applied to one or more intermediate layers **1120**. The chip **1124** may be a system-on-a-chip (SoC), an application-specific integrated circuit (ASIC), and/or the like that includes circuitry for a contacted payment interface and/or a wireless payment interface for a POS terminal.

In the present embodiment, the chip **1124** comprises both the circuitry for contacted and wireless payment interfaces. The contacted payment interface may utilize the processor **1210**, a memory **1220**, a communication interface **1240**, and contacts **1230** via, e.g., the dashed circuit connections. For instance, when the payment instrument **1000** is inserted into a POS terminal, the contacts **1230** electrically connect with the POS terminal, providing power to the processor **1210**, the memory **1220**, and the communications interface **1240**. The communications interface **1240** may receive a communication via the contacts **1230** and provide at least part of the communication to the processor **1210**. The processor **1210** may access the memory **1220** to respond to the communication and cause the communication interface **1240** to respond to the communication via the contacts **1230**.

The wireless payment interface may utilize the processor **1210**, the memory **1220**, the communication interface **1240**, a power converter **1250**, and an energy storage **1260**. For instance, when brought near a NFC radio in the POS terminal, the antenna **1130** may receive energy transmitted from the NFC radio and collect the energy in the energy storage **1260**, once the voltage on the energy storage reaches a minimum voltage level for operation of the processor **1210**, the memory **1220**, and the communication interface **1240**, communication interface **1240** may receive a communication from the NFC radio of the POS terminal, and the processor **1210** may access the memory **1220** to prepare a response. Once the energy storage **1260** has stored enough energy measured, e.g., by the voltage, to power the wireless transmitter of the communication interface **1240**, the processor **1210** may cause the wireless transmitter of the communication interface **1240** to wirelessly transmit the response via the antenna **1130** to the NFC radio.

The power converter **1250** may store the energy from the NFC radio transmission in the energy storage **1260**, measure the voltage on the energy storage **1260**, and supply power to the wireless transmitter to transmit the response via the antenna. In other embodiments, the power converter **1250** may convert the voltage of the energy prior to storage in the energy storage **1260**. In some embodiments, the energy storage **1260** may comprise a capacitance-based device to store the energy. In other embodiments, the payment instrument **1000** may include a battery for energy storage **1260**.

FIGS. **2**A-C depict alternative embodiments of intermediate layers, such as the intermediate layers **1020** and **1120** illustrated in FIGS. **1**A-**1**C. FIG. **2**A depicts an alternative embodiment of an intermediate layer **2020** of a payment instrument **2000**, such as the intermediate layer **1020** illustrated in FIG. **1**A. The payment instrument **2000** may comprise layers of plastic, metal, and/or other material. In the present embodiment, the payment instrument **2000** comprises an intermediate layer **2020**. The intermediate layer **2020** may comprise an absorbent medium **2010**, reservoir(s) **1122**, and hidden element(s) **2030**. The hidden element(s) **2030** may comprise codes, logos, other graphics, words, numbers, and/or the like. In some embodiments, the hidden element(s) **2030** may provide fraud protection to help identify counterfeit payment instruments. In some embodiments, the hidden element(s) **2030** may include a security feature that can be scanned, the hidden element(s) **2030** may obscure payment instrument numbers on the face of the payment instrument as the absorbent medium **1126** absorbs dyes from the reservoir(s) **1122**, the hidden element(s) **2030** may include one or more three-dimensional (3D) features that are not present in a two-dimensional image of the payment instrument, and/or the like. The **3**D features may include features that refract light at different angles to provide one or more different images that change over time in response to the absorption and curing of the dyes released from the reservoir(s) **1122**. For instance, the cloth of the absorbent medium **2010** may include glass flakes or particles, metallic flakes or particles, and/or the like to create reflective surfaces.

FIG. **2**B depicts an alternative embodiment of an intermediate layer **2120**, such as the one or more intermediate layers **1020** illustrated in FIG. **1**A. The intermediate layer **2120** comprises an absorbent medium **2110** such as a cloth medium or other medium and multiple reservoir(s) **1122**. In the present embodiment, the reservoir(s) **1122** comprise limiting valves **2130**. The limiting valves **2130** may limit the rate of dye released onto the absorbent medium **2110** in response to pressure on the reservoir(s) **1122** from, e.g., inserting the payment instrument **2100** into a POS terminal or responsive to pressure applied by a consumer.

FIG. **2**B also illustrates a translucent window **1050** of the top layer of the payment instrument **2100**. The translucent window **1050** may be any shape, may include a tint of any one or more colors, may not include a tint, may be part of a security feature to detect counterfeit payment instruments, and/or the like.

FIG. **2**C depicts an alternative embodiment of an intermediate layer **2220**, such as the one or more intermediate layers **1020** illustrated in FIG. **1**A. The intermediate layer **2220** comprises a translucent window **1050** medium **2210** and multiple reservoir(s) **1122** proximate to the chip **1124**. In the present embodiment, the reservoir(s) **1122** are hidden by opaque portions of the intermediate layer **2220** and/or opaque portions of the top layer **1030**. The present embodiment may also include curved channels **2212** and **2214** to disperse dyes from the reservoir(s) **1022**A and **1022**B over different areas of the surface of the absorbent medium **2210**.

In some embodiments, the channels **2212** and **2214** may reside in different layers of the payment instrument **2200** such as the bottom layer and/or one or more of the intermediate layers. In other embodiments, the channels **2212** and **2214** may reside in the same layer of the payment instrument **2200**. In some embodiments, the pattern of the channels **2212** and **2214** may be selectable at the time of manufacture of the payment instrument **2200**. For instance, multiple patterns of the channels **2212** and **2214** may be available and the pattern of the channels may be selected randomly at the time of manufacture of the payment instrument **2200**.

FIG. **2**D depicts an embodiment of the reservoir(s) **1122** of a payment instrument **2300**, such as the payment instruments illustrated in FIGS. **1**A-C and **2**A-C. The reservoir(s) **1122** comprise a dye in the form of beads **2310** that dissolve and/or can be ruptured through use or processing of the payment instrument **2300**. In the present embodiment, the beads have a circular **2**D profile and are spherical, but the beads can be any shape. In some embodiments, processing the payment instrument may involve application of heat and/or pressure to the payment instrument to cause the beads to dissolve and/or rupture to release dyes within the beads.

FIGS. **3**A-D depict embodiments of flowcharts for processes to manufacture payment instruments such as the payment instruments shown in FIGS. **1**A-B and **2**A-D. FIG. **3**A illustrates a flowchart for a process **3000** for a payment instrument with a unique design. The process **3000** starts with provision of a bottom layer (element **3010**). In some embodiments, one or more reservoirs may be created in the bottom layer as cavities filled with one or more different color dyes in a form of, e.g., beads. Some of these embodiments may etch one or more channels or otherwise include channels in the bottom layer to direct a flow of the dyes onto an absorption medium.

After preparing the bottom layer, one or more intermediate layers are attached to the bottom layer, wherein at least one of the one or more intermediate layers comprises: a chip comprising a processor, memory, and a communication interface; an antenna coupled with the chip; an absorption medium; and, in some embodiments, one or more reservoirs comprising beads (element **3015**). The beads may comprise the dyes and may be configured to release the dyes over time onto the absorption medium. In some embodiments, the colors of dyes in the beads may be selected randomly and one or more colors of dyes may be included in the form of beads in the same reservoir to create a single blended color or an evolving color as the colors absorb into the absorption medium. The dyes may cure in response to a curing agent applied to the absorption medium and/or included in beads in the one or more reservoir(s) to generate a unique design on the absorption medium.

Once the intermediate layers are attached to the bottom layer, a top layer is attached to the intermediate layer(s) to form the payment instrument (element **3020**). The top layer may have at least a partially translucent portion above the absorption medium. The absorption medium in conjunction with the dyes as released from the reservoirs forms a unique design based on curing times, base colors in the absorption medium, combinations of colors that absorb into the same portions of the absorption medium, and/or the like. The unique design of the absorption medium is at least partially visible through the at least one partially translucent portion.

In some embodiments, the beads are configured to break in response to placement of the chip into a POS terminal. In some embodiments, the beads may also or alternatively dissolve over time. In several embodiments, the absorption medium may comprise a curing agent that varies in concentration across the span of the absorption medium and the beads may comprise a dye to color the absorption medium. In some embodiments, the beads may dissolve in the presence of dye or another agent in the reservoir that is external to the beads.

In some embodiments, the absorption medium comprises a security feature to distinguish counterfeit payment instruments. For instance, one or more 2D or 3D features may be included in the absorption medium through preparation of a portion of the surface of the absorption medium to cause the one or more 2D or 3D features to become visible in the presence of cured or curing dyes. The 2D or 3D features may form codes, text, numbers, logos, other graphics, and/or the like. In some embodiments, the security feature may appear different from perspectives or points of view. For example, a code, text, number, or the like may change from a different perspective.

In some embodiments, the absorption medium may include a coating over a portion of a surface of the absorption medium to prevent the surface from absorbing to the dyes. In some embodiments, the absorption medium may include a coating over a portion of a surface of the absorption medium to slow the curing of the dyes. In some embodiments, the absorption medium may include a coating over a portion of a surface of the absorption medium to hasten the curing of the dyes. Some embodiments may include beads with different dye colors having the same or different absorption rates in the same or in different reservoirs and, in some embodiments, the beads with the different dye colors or agents may dissolve at different rates. For example, a first set of beads may include a yellow dye and may dissolve at a first rate. A second set of beads may include a red dye and may dissolve at a faster rate than the first rate and a third set of beads may include a blue dye and may dissolve at a slower rate than the first rate. Furthermore, a fourth set of beads may include a concentrated curing agent to increase the curing rate of the dyes proximal to the fourth set of beads.

In some embodiments, the absorbent medium has a base color and a first color dye may combine with the base color to create a different color. In some embodiments, the absorbent medium has a base color and a second color dye may cure as a translucent layer over the base color to create a different color. In further embodiments, the absorbent medium may include more than one type of metal such as metal foil, metal flakes, metal particles, and/or the like.

In some embodiments, the top layer comprises conductive contacts to electrically connect the chip with a point of sale (POS) terminal, wherein at least one reservoir of the one or more reservoirs resides in proximity to the chip to insert the at least one reservoir at least partially into a POS terminal to create an electrical connection between the conductive contacts and the POS terminal.

FIG. **3**B illustrates a flowchart for a process **3100** for a payment instrument with a unique design. The process **3100** starts with provision of a bottom layer (element **3110**). Some embodiments may laminate the bottom layer with the one or more intermediate layers to the bottom layer and the one or more intermediate layers may comprise: a chip comprising a processor, memory, and a communication interface; an antenna coupled with the chip; an absorption medium; and one or more reservoirs comprising beads with one or more different colors of dyes (element **3115**). The reservoirs may interconnect with the absorption medium via limit valves configured to limit the rate of release of the dyes onto the absorption medium. The dyes may cure at a curing rate based on the curing agent on the absorption medium. In some embodiments, some of the beads may include white dyes to vary the spectrum of the color of one or more different color dyes as the dyes are absorbed into the absorption medium.

After laminating the bottom layer with the intermediate layers, a top layer is laminated over the intermediate layer(s) to form the payment instrument (element **3120**). The top layer may have at least a partially translucent portion above the absorption medium. The absorption medium in conjunction with the dyes released from the reservoirs form a unique design such as a tie-dye design visible via a translucent window in the top layer of the payment instrument. The unique design of the absorption medium is at least partially visible through the at least one partially translucent window in the top layer of the payment instrument. In some embodiments, the at least partially translucent portion is transparent and, in some embodiments, the at least partially translucent portion is tinted.

FIG. **3**C illustrates a flowchart for a process **3200** for a payment instrument with a unique design. The process **3200** starts with provision of a bottom layer (element **3210**). Some embodiments may attach one or more intermediate layers to the bottom layer (element **3215**). At least one of the one or more intermediate layers may comprise: a chip comprising a processor, memory, and a communication interface; an antenna coupled with the chip; an absorbent medium; and two or more reservoirs comprising beads. A first reservoir of the two or more reservoirs may comprise beads having a first set of one or more different color dyes. A second reservoir of the two or more reservoirs may comprise beads having a second set of one or more different color dyes. The beads may be configured to release the dyes onto the absorbent medium and the dyes may cure on the absorbent medium in response to a curing agent included on the absorbent medium or in other beads within the reservoirs.

In some embodiments, the one or more intermediate layers may comprise a set of channels, selected from more than one sets of channels, to interconnect the two or more reservoirs with the absorbent medium to direct dyes from the two or more reservoirs onto the absorbent medium. The reservoirs may interconnect with the absorption medium via channels configured to release the dyes onto the absorbent medium in a pattern.

After laminating the bottom layer with the intermediate layers, a top layer is laminated over the intermediate layer(s) to form the payment instrument (element **3220**). The top layer may have at least a partially translucent portion or window above the absorbent medium. The absorbent medium may be at least partially visible through the at least one partially translucent portion.

In some embodiments, after attaching the top layer to the payment instrument, the process **3200** may process the payment instrument to burst and/or dissolve at least some of the beads in at least one of the two or more reservoirs (element **3225**).

FIG. **4** illustrates an embodiment of a system **4000**. The system **4000** is a computer system with multiple processor cores such as a distributed computing system, supercomputer, high-performance computing system, computing cluster, mainframe computer, mini-computer, client-server system, personal computer (PC), workstation, server, portable computer, laptop computer, tablet computer, handheld device such as a personal digital assistant (PDA), or other device for processing, displaying, or transmitting information. Similar embodiments may comprise, e.g., entertainment devices such as a portable music player or a portable video player, a smart phone or other cellular phone, a telephone, a digital video camera, a digital still camera, an external storage device, or the like. Further embodiments implement larger scale server configurations. In other embodiments, the system **4000** may have a single processor with one core or more than one processor. Note that the term "processor" refers to a processor with a single core or a processor package with multiple processor cores.

As shown in FIG. **4**, system **4000** comprises a motherboard **4005** for mounting platform components. The motherboard **4005** is a point-to-point interconnect platform that includes a first processor **4010** and a second processor **4030** coupled via a point-to-point interconnect **4056** such as an Ultra Path Interconnect (UPI). In other embodiments, the system **4000** may be of another bus architecture, such as a multi-drop bus. Furthermore, each of processors **4010** and **4030** may be processor packages with multiple processor cores including processor core(s) **4020** and **4040**, respectively. While the system **4000** is an example of a two-socket (2S) platform, other embodiments may include more than two sockets or one socket. For example, some embodiments may include a four-socket (4S) platform or an eight-socket (8S) platform. Each socket is a mount for a processor and may have a socket identifier. Note that the term platform refers to the motherboard with certain components mounted such as the processors **4010** and the chipset **4060**. Some platforms may include additional components and some platforms may only include sockets to mount the processors and/or the chipset.

The first processor **4010** includes an integrated memory controller (IMC) **4014** and point-to-point (P-P) interconnects **4018** and **4052**. Similarly, the second processor **4030** includes an IMC **4034** and P-P interconnects **4038** and **4054**. The IMC's **4014** and **4034** couple the processors **4010** and **4030**, respectively, to respective memories, a memory **4012** and a memory **4032**. The memories **4012** and **4032** may be portions of the main memory (e.g., a dynamic random-access memory (DRAM)) for the platform such as double data rate type 3 (DDR3) or type 4 (DDR4) synchronous DRAM (SDRAM). In the present embodiment, the memories **4012** and **4032** locally attach to the respective processors **4010** and **4030**. In other embodiments, the main memory may couple with the processors via a bus and shared memory hub.

The processors **4010** and **4030** comprise caches coupled with each of the processor core(s) **4020** and **4040**, respectively. In the present embodiment, the processor core(s) **4020** of the processor **4010** include a logic circuitry **4026** such as the processes described in conjunction with FIGS. **3**A-C. The logic circuitry **4026** may represent circuitry configured to implement the functionality to manufacture a payment instrument within the processor core(s) **4020** or may represent a combination of the circuitry within a processor and a medium (such as the logic circuitry **1225** shown in FIG. **1**B) to store all or part of the functionality of the logic circuitry **4026** in memory such as cache, the memory **4012**, buffers, registers, and/or the like. In several embodiments, the functionality of the logic circuitry **4026** resides in whole or in part as code in a memory such as the lock logic circuitry **4096** in the data storage unit **4088** attached to the processor **4010** via a chipset **4060** such as the lock logic circuitry **1225** shown in FIG. **1**B. The functionality of the logic circuitry **4026** may also reside in whole or in part in memory such as the memory **4012** and/or a cache of the processor. Furthermore, the functionality of the logic circuitry **4026** may also reside in whole or in part as circuitry within the processor **4010** and/or **4030** and may perform operations, e.g., within registers or buffers such as the registers **4016** and **4036** within the processors **4010** and **4030**, respectively, or within an instruction pipeline of the processor **4010** and **4030**, respectively.

In other embodiments, more than one of the processors **4010** and **4030** may comprise the functionality of the logic circuitry **4026** such as the processor **4030** and/or the processor within the deep learning accelerator **4067** coupled with the chipset **4060** via an interface (I/F) **4066**. The I/F **4066** may be, for example, a Peripheral Component Interconnect-enhanced (PCI-e).

The first processor **4010** couples to a chipset **4060** via P-P interconnects **4052** and **4062** and the second processor **4030** couples to a chipset **4060** via P-P interconnects **4054** and **4064**. Direct Media Interfaces (DMIs) **4057** and **4058** may couple the P-P interconnects **4052** and **4062** and the P-P interconnects **4054** and **4064**, respectively. The DMI may be a high-speed interconnect that facilitates, e.g., eight Giga Transfers per second (GT/s) such as DMI **3.0**. In other embodiments, the processors **4010** and **4030** may interconnect via a bus.

The chipset **4060** may comprise a controller hub such as a platform controller hub (PCH). The chipset **4060** may include a system clock to perform clocking functions and include interfaces for an I/O bus such as a universal serial bus (USB), peripheral component interconnects (PCIs), serial peripheral interconnects (SPIs), integrated interconnects (I2Cs), and the like, to facilitate connection of peripheral devices on the platform. In other embodiments, the chipset **4060** may comprise more than one controller hub such as a chipset with a memory controller hub, a graphics controller hub, and an input/output (I/O) controller hub.

In the present embodiment, the chipset **4060** couples with a trusted platform module (TPM) **4072** and the unified extensible firmware interface (UEFI), BIOS, Flash component **4074** via an interface (I/F) **4070**. The TPM **4072** is a dedicated microcontroller designed to secure hardware by integrating cryptographic keys into devices. The UEFI, BIOS, Flash component **4074** may provide pre-boot code.

Furthermore, chipset **4060** includes an I/F **4066** to couple chipset **4060** with a high-performance graphics engine, graphics card **4065**. In other embodiments, the system **4000** may include a flexible display interface (FDI) between the processors **4010** and **4030** and the chipset **4060**. The FDI interconnects a graphics processor core in a processor with the chipset **4060**.

Various I/O devices **4092** couple to the bus **4081**, along with a bus bridge **4080** which couples the bus **4081** to a second bus **4091** and an I/F **4068** that connects the bus **4081** with the chipset **4060**. In one embodiment, the second bus **4091** may be a low pin count (LPC) bus. Various devices may couple to the second bus **4091** including, for example, a keyboard **4082**, a mouse **4084**, communication devices **4086** and a data storage unit **4088** that may store code such as the logic circuitry **4096**. Furthermore, an audio I/O **4090** may couple to second bus **4091**. Many of the I/O devices **4092**, communication devices **4086**, and the data storage unit **4088** may reside on the motherboard **4005** while the keyboard **4082** and the mouse **4084** may be add-on peripherals. In other embodiments, some or all the I/O devices **4092**, communication devices **4086**, and the data storage unit **4088** are add-on peripherals and do not reside on the motherboard **4005**.

FIG. **5** illustrates an example of a storage medium **5000** to manufacture a payment instrument. Storage medium **5000** may comprise an article of manufacture. In some examples, storage medium **5000** may include any non-transitory computer readable medium or machine readable medium, such as an optical, magnetic or semiconductor storage. Storage medium **5000** may store various types of computer executable instructions, such as instructions to implement logic flows and/or techniques described herein. Examples of a computer readable or machine-readable storage medium may include any tangible media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writeable memory, and so forth. Examples of computer executable instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, object-oriented code, visual code, and the like. The examples are not limited in this context.

FIG. **6** illustrates an example computing platform **6000**. In some examples, as shown in FIG. **6**, computing platform **6000** may include a processing component **6010**, other platform components or a communications interface **6030**. According to some examples, computing platform **6000** may be implemented in a computing device such as a server in a system such as a data center or server farm that supports a manager or controller for managing configurable computing resources as mentioned above. Furthermore, the communications interface **6030** may comprise a wake-up radio (WUR) and may be capable of waking up a main radio of the computing platform **6000**.

According to some examples, processing component **6010** may execute processing operations or logic for apparatus **6015** described herein such as the processes described in conjunction with FIGS. **3**A-C. Processing component **6010** may include various hardware elements, software elements, or a combination of both. Examples of hardware elements may include devices, logic devices, components, processors, microprocessors, circuits, processor circuits, circuit elements (e.g., transistors, resistors, capacitors, inductors, and so forth), integrated circuits, application specific integrated circuits (ASIC), programmable logic devices (PLD), digital signal processors (DSP), field programmable gate array (FPGA), memory units, logic gates, registers, semiconductor device, chips, microchips, chip sets, and so forth. Examples of software elements, which may reside in the storage medium **6020**, may include software components, programs, applications, computer programs, application programs, device drivers, system programs, software development programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, functions, methods, procedures, software interfaces, application program interfaces (API), instruction sets, computing code, computer code, code segments, computer code segments, words, values, symbols, or any combination thereof. Determining whether an example is implemented using hardware elements and/or software elements may vary in accordance with any number of factors, such as desired computational rate, power levels, heat tolerances, processing cycle budget, input data rates, output data rates, memory resources, data bus speeds and other design or performance constraints, as desired for a given example.

In some examples, other platform components **6025** may include common computing elements, such as one or more processors, multi-core processors, co-processors, memory units, chipsets, controllers, peripherals, interfaces, oscillators, timing devices, video cards, audio cards, multimedia input/output (I/O) components (e.g., digital displays), power supplies, and so forth. Examples of memory units may include without limitation various types of computer readable and machine readable storage media in the form of one or more higher speed memory units, such as read-only memory (ROM), random-access memory (RAM), dynamic RAM (DRAM), Double-Data-Rate DRAM (DDRAM), synchronous DRAM (SDRAM), static RAM (SRAM), programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), flash memory, polymer memory such as ferroelectric polymer memory, ovonic memory, phase change or ferroelectric memory, silicon-oxide-nitride-oxide-silicon (SONOS) memory, magnetic or optical cards, an array of devices such as Redundant Array of Independent Disks (RAID) drives, solid state memory devices (e.g., USB memory), solid state drives (SSD) and any other type of storage media suitable for storing information.

In some examples, communications interface **6030** may include logic and/or features to support a communication interface. For these examples, communications interface **6030** may include one or more communication interfaces that operate according to various communication protocols or standards to communicate over direct or network communication links. Direct communications may occur via use of communication protocols or standards described in one or more industry standards (including progenies and variants) such as those associated with the PCI Express specification. Network communications may occur via use of communication protocols or standards such as those described in one or more Ethernet standards promulgated by the Institute of Electrical and Electronics Engineers (IEEE). For example, one such Ethernet standard may include IEEE 802.3-2012, Carrier sense Multiple access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications, Published in December 2012 (hereinafter "IEEE 802.3"). Network communication may also occur according to one or more OpenFlow specifications such as the OpenFlow Hardware Abstraction API Specification. Network communications may also occur according to Infiniband Architecture Specification, Volume 1, Release 1.3, published in March 2015 ("the Infiniband Architecture specification").

Computing platform **6000** may be part of a computing device that may be, for example, a server, a server array or server farm, a web server, a network server, an Internet server, a workstation, a mini-computer, a main frame computer, a supercomputer, a network appliance, a web appliance, a distributed computing system, multiprocessor systems, processor-based systems, or combination thereof. Accordingly, functions and/or specific configurations of computing platform **6000** described herein, may be included, or omitted in various embodiments of computing platform **6000**, as suitably desired.

The components and features of computing platform **6000** may be implemented using any combination of discrete circuitry, ASICs, logic gates and/or single chip architectures. Further, the features of computing platform **6000** may be implemented using microcontrollers, programmable logic arrays and/or microprocessors or any combination of the foregoing where suitably appropriate. It is noted that hardware, firmware and/or software elements may be collectively or individually referred to herein as "logic".

It should be appreciated that the exemplary computing platform **6000** shown in the block diagram of FIG. **6** may represent one functionally descriptive example of many potential implementations. Accordingly, division, omission or inclusion of block functions depicted in the accompanying figures does not infer that the hardware components, circuits, software and/or elements for implementing these functions would necessarily be divided, omitted, or included in embodiments.

One or more aspects of at least one example may be implemented by representative instructions stored on at least one machine-readable medium which represents various logic within the processor, which when read by a machine, computing device or system causes the machine, computing device or system to fabricate logic to perform the techniques described herein. Such representations, known as "IP cores", may be stored on a tangible, machine readable medium and supplied to various customers or manufacturing facilities to load into the fabrication machines that make the logic or processor.

Various examples may be implemented using hardware elements, software elements, or a combination of both. In some examples, hardware elements may include devices, components, processors, microprocessors, circuits, circuit elements (e.g., transistors, resistors, capacitors, inductors, and so forth), integrated circuits, application specific integrated circuits (ASIC), programmable logic devices (PLD), digital signal processors (DSP), field programmable gate array (FPGA), memory units, logic gates, registers, semiconductor device, chips, microchips, chip sets, and so forth. In some examples, software elements may include software components, programs, applications, computer programs, application programs, system programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, functions, methods, procedures, software interfaces, application program interfaces (API), instruction sets, computing code, computer code, code segments, computer code segments, words, values, symbols, or any combination thereof. Determining whether an example is implemented using hardware elements and/or software elements may vary in accordance with any number of factors, such as desired computational rate, power levels, heat tolerances, processing cycle budget, input data rates, output data rates, memory resources, data bus speeds and other design or performance constraints, as desired for a given implementation.

Some examples may include an article of manufacture or at least one computer-readable medium. A computer-readable medium may include a non-transitory storage medium to store logic. In some examples, the non-transitory storage medium may include one or more types of computer-readable storage media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writeable memory, and so forth. In some examples, the logic may include various software elements, such as software components, programs, applications, computer programs, application programs, system programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, functions, methods, procedures, software interfaces, API, instruction sets, computing code, computer code, code segments, computer code segments, words, values, symbols, or any combination thereof.

According to some examples, a computer-readable medium may include a non-transitory storage medium to store or maintain instructions that when executed by a machine, computing device or system, cause the machine, computing device or system to perform methods and/or operations in accordance with the described examples. The instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. The instructions may be implemented according to a predefined computer language, manner, or syntax, for instructing a machine, computing device or system to perform a certain function. The instructions may be implemented using any suitable high-level, low-level, object-oriented, visual, compiled and/or interpreted programming language.

Some examples may be described using the expression "in one example" or "an example" along with their derivatives. These terms mean that a particular feature, structure, or characteristic described in connection with the example is included in at least one example. The appearances of the phrase "in one example" in various places in the specification are not necessarily all referring to the same example.

Some examples may be described using the expression "coupled" and "connected" along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, descriptions using the terms "connected" and/or "coupled" may indicate that two or more elements are in direct physical or electrical contact with each other. The term "coupled," however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

In addition, in the foregoing Detailed Description, various features are grouped together in a single example for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate example. In the appended claims, the terms "including" and "in which" are used as the plain-English equivalents of the respective terms "comprising" and "wherein," respectively. Moreover, the terms "first," "second," "third," and so forth, are used merely as labels, and are not intended to impose numerical requirements on their objects.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

A data processing system suitable for storing and/or executing program code will include at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories which provide temporary storage of at least some program code to reduce the number of times code must be retrieved from bulk storage during execution. The term "code" covers a broad range of software components and constructs, including applications, drivers, processes, routines, methods, modules, firmware, microcode, and subprograms. Thus, the term "code" may be used to refer to any collection of instructions which, when executed by a processing system, perform a desired operation or operations.

Logic circuitry, devices, and interfaces herein described may perform functions implemented in hardware and also implemented with code executed on one or more processors. Logic circuitry refers to the hardware or the hardware and code that implements one or more logical functions. Circuitry is hardware and may refer to one or more circuits. Each circuit may perform a particular function. A circuit of the circuitry may comprise discrete electrical components interconnected with one or more conductors, an integrated circuit, a chip package, a chip set, memory, or the like. Integrated circuits include circuits created on a substrate such as a silicon wafer and may comprise components. And integrated circuits, processor packages, chip packages, and chipsets may comprise one or more processors.

Processors may receive signals such as instructions and/or data at the input(s) and process the signals to generate the at least one output. While executing code, the code changes the physical states and characteristics of transistors that make up a processor pipeline. The physical states of the transistors translate into logical bits of ones and zeros stored in registers within the processor. The processor can transfer the physical states of the transistors into registers and transfer the physical states of the transistors to another storage medium.

A processor may comprise circuits to perform one or more sub-functions implemented to perform the overall function of the processor. One example of a processor is a state machine or an application-specific integrated circuit (ASIC) that includes at least one input and at least one output. A state machine may manipulate the at least one input to generate the at least one output by performing a predetermined series of serial and/or parallel manipulations or transformations on the at least one input.

The logic as described above may be part of the design for an integrated circuit chip. The chip design is created in a graphical computer programming language and stored in a computer storage medium or data storage medium (such as a disk, tape, physical hard drive, or virtual hard drive such as in a storage access network). If the designer does not fabricate chips or the photolithographic masks used to fabricate chips, the designer transmits the resulting design by physical means (e.g., by providing a copy of the storage medium storing the design) or electronically (e.g., through the Internet) to such entities, directly or indirectly. The stored design is then converted into the appropriate format (e.g., GDSII) for the fabrication.

The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case, the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher-level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case, the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either (a) an intermediate product, such as a processor board, a server platform, or a motherboard, or (b) an end product.

What is claimed is:

1. A payment instrument comprising:

a bottom layer;

one or more intermediate layers, wherein at least one of the one or more intermediate layers comprises:

a chip;

an absorbent medium comprising a curing agent; and

one or more reservoirs, wherein a first reservoir of the one or more reservoirs comprises a first color dye and a second reservoir of the one or more reservoirs a second color dye, configured to release the dyes onto the absorbent medium, the dyes to cure within the absorbent medium; and

a top layer, wherein the absorbent medium is at least partially visible through the top layer.

2. The payment instrument of claim **1**, conductive contacts of the chip to connect with a point of sale (POS) terminal, wherein at least one reservoir of the one or more reservoirs resides in proximity to the chip to insert the at least one reservoir at least partially into the POS terminal to create an electrical connection between the conductive contacts and the POS terminal.

3. The payment instrument of claim **2**, wherein the dyes reside in beads that are configured to break in response to placement of the chip into a POS terminal.

4. The payment instrument of claim **3**, wherein the beads dissolve over time or dissolve in a presence of the dyes from burst beads.

5. The payment instrument of claim **1**, wherein the absorbent medium comprises a first base color, the first color dye changes the absorbent medium to a first color and the second color dye changes the absorbent medium to a second color.

6. The payment instrument of claim **5**, wherein the absorbent medium comprises a security feature to distinguish counterfeit payment instruments.

7. The payment instrument of claim **5**, wherein the absorbent medium includes a coating over a portion of a surface of the absorbent medium to prevent the absorbent medium from absorbing the dyes.

8. The payment instrument of claim **5**, wherein the absorbent medium comprises a second base color and the first color dye combines with the second base color to create a third color.

9. The payment instrument of claim **8**, wherein the second color dye cures as a translucent layer of the second color over the second base color to create a fourth color.

10. A payment instrument comprising:

a bottom layer;

one or more intermediate layers, wherein at least one of the one or more intermediate layers comprises:

a chip;

an absorbent medium; and

a first set of beads comprising a first color dye;

a second set of beads comprising a second color dye, wherein the first set of beads and the second set of beads are configured to release the dyes onto the absorbent medium, the dyes to cure on the absorbent medium; and

a top layer, wherein the absorbent medium is at least partially visible through the top layer.

11. The payment instrument of claim **10**, wherein the first set of beads and the second set of beads comprise dyes that cure at different rates.

12. The payment instrument of claim **10**, in the first set of beads and the second set of beads dissolve over time.

13. The payment instrument of claim **10**, wherein a third set of beads having one or more different color dyes are randomly included in one or more reservoirs or cavities.

14. The payment instrument of claim **10**, wherein the one or more intermediate layers include a channel coupled to at least one or more reservoirs, the one or more reservoirs to contain beads from the first set, the second set, or a combination thereof, to direct dye from the at least one or more reservoirs onto the absorbent medium.

15. The payment instrument of claim **14**, wherein the absorbent medium comprises a first base color, the first color dye changes the first base color to a first color and the second color dye changes the first base color to a second color.

16. The payment instrument of claim **10**, wherein the absorbent medium comprises a three-dimensional security feature to distinguish counterfeit payment instruments.

17. The payment instrument of claim **10**, wherein the absorbent medium includes a coating over a portion of a surface of the absorbent medium to prevent the absorbent medium from absorbing the dyes.

18. The payment instrument of claim **10**, wherein the absorbent medium comprises a second base color and the first color dye combines with the second base color to create a third color.

19. The payment instrument of claim **10**, wherein the absorbent medium comprises a fourth base color and the second color dye applies a translucent layer of a second color over the fourth base color to create a fourth color.

20. A method to create a payment instrument, the method comprising:

providing a bottom layer;

attaching one or more intermediate layers, wherein at least one of the one or more intermediate layers comprises:

a chip;

an absorbent medium;

one or more reservoirs, wherein a first reservoir of the one or more reservoirs comprises a first set of one or more different color dyes and a second reservoir of the one or more reservoirs comprises a second set of one or more different color dyes, configured to release the dyes onto the absorbent medium, the dyes to cure on the absorbent medium; and

a set of channels, selected from more than one sets of channels, to interconnect the one or more reservoirs with the absorbent medium to direct dyes from the one or more reservoirs onto the absorbent medium; and

attaching a top layer, wherein at least part of a surface of the absorbent medium is visible through the top layer.

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